School Sphere - External Services & Subscriptions

Static review of SMS UI, all backend services and SMS Mobile. Generated 2026-11-04.

Required to run in production

Service	Used for	Config
Azure SQL (SQL Server)	One database per service. Local Docker uses a SQL Server container.	AZURE_SQL_*
Azure Blob Storage	All uploaded files: study materials, photos, documents, talent media, meeting files, avatars. Locally: Azurite.	AZURE_STORAGE_CONNECTION_STRING
VM / hosting + DNS	Domain A record, www, wildcard *.sms-schoolsphere.com for school subdomains.	DOMAIN, APP_DOMAIN, TENANT_BASE_DOMAIN
Let's Encrypt	Automatic certificates via Caddy, issued per school on first visit.	ACME_EMAIL
GitHub + GHCR	CI builds and container registry.	GHCR_REGISTRY, docker-publish.yml
Redis	AuthService cache (self-hosted).	ConnectionStrings__Redis
LiveKit	In-app video classes (self-hosted) plus Egress recording; needs open ports and 2-4 vCPUs per recording.	LIVEKIT_*
Brevo SMTP	System email: password resets, landing-page leads, platform default for communication email.	SMTP_*

Optional / switchable (off by default)

Service	Used for	Default
Anthropic / OpenAI / Azure OpenAI	AI assistant, generators, report remarks	Mock
OpenAI / Azure OpenAI embeddings	Study-material search	Mock
Razorpay	Online fee payment on web and mobile apps	Off when keys empty
Stripe, Cashfree, PayU	Per-school payment gateways (each school saves its own encrypted keys)	Off
MSG91 / Twilio	OTP sign-in SMS (requires OTP_ENABLED=true)	Logging
MSG91 / Twilio / Brevo API / Meta WhatsApp Cloud API	Communication Center SMS, email, WhatsApp (each school connects its own account)	Per school

Web Push (VAPID)	Browser push notifications; no vendor, keys generated by you	Off
Expo Push service	Mobile app push notifications	Used by mobile apps
CompreFace	Face-recognition attendance (self-hosted, needs consent)	None
Local LLM (Ollama)	Optional AI provider	-

Client-side third parties

- Google Analytics (gtag) - hardcoded ID G-H52D8T9BX8
- Google Fonts - Inter and talents CSS
- OpenStreetMap tiles - transport maps and stop picker (free, tile usage policy limits heavy use)
- YouTube (nocookie) - embedded videos in study materials
- Razorpay and Cashfree checkout SDKs from their CDNs

Mobile apps (SMS Mobile)

- Sentry - crash reporting (EXPO_PUBLIC_SENTRY_DSN)
- Expo / EAS - builds and push
- Apple Developer and Google Play accounts for store releases
- Firebase/FCM and APNs - reached through Expo push, not direct SDKs
- LiveKit client and react-native-razorpay

Leftovers

- amplify/ (AWS Amplify) is the stock scaffold; nothing from AWS is actually used.
- WhatsApp notification for landing-page leads (LEAD_NOTIFY_WHATSAPP) is only a logging stub.
- SMTP_USERNAME defaults to a real Brevo login in compose files; consider moving it into .env.

Minimum paid set for a real launch

- Azure SQL, Azure Blob Storage, a VM with domain and wildcard DNS, Brevo
- An SMS provider (MSG91 or Twilio) if OTP is enabled
- An AI provider key if real AI is wanted instead of Mock
- Razorpay if online fee payment is wanted
- Apple and Google developer accounts, and Sentry, for the mobile apps